ANN Viva Questions & Short Answers

1. What is an Artificial Neural Network?

ANN is a computational model inspired by the human brain used for learning and prediction.

2. Why are ANNs inspired by the human brain?

Because ANN works using interconnected neurons similar to biological neurons.

3. What are the main components of a neuron?

Inputs, weights, bias, activation function, and output.

4. What is the role of weights and bias in ANN?

Weights control input importance and bias shifts output.

5. What is an activation function?

It adds non-linearity and decides neuron activation.

6. Why do we need activation functions in neural networks?

To solve complex non-linear problems.

7. Differentiate between biological neuron and artificial neuron.

Biological neuron is natural; artificial neuron is mathematical model.

8. What is supervised learning?

Learning using labeled data.

9. What is unsupervised learning?

Learning using unlabeled data.

10. What is reinforcement learning?

Learning through rewards and penalties.

11. What is a single-layer perceptron?

A neural network with only one layer.

12. What is a multilayer perceptron (MLP)?

ANN with multiple hidden layers.

13. What are input, hidden, and output layers?

Input takes data, hidden processes it, output gives result.

14. Why are hidden layers important?

They help learn complex patterns.

15. What is deep learning?

Deep learning uses many hidden layers.

16. Differentiate between shallow and deep neural networks.

Shallow has few layers; deep has many layers.

17. What is the purpose of neurons in hidden layers?

To extract features and patterns.

18. What is a feedforward neural network?

Data moves only from input to output.

19. What is recurrent feedback in neural networks?

Output is fed back into the network.

20. What are the limitations of a single-layer perceptron?

Cannot solve non-linear problems like XOR.

21. Explain the sigmoid activation function.

Sigmoid gives output between 0 and 1.

22. What are the advantages and disadvantages of sigmoid?

Advantage: smooth output. Disadvantage: vanishing gradient.

23. What is the ReLU activation function?

Returns 0 for negative input and same value for positive.

24. Why is ReLU widely used?

It is fast and reduces vanishing gradient.

25. What is the tanh activation function?

Gives output between -1 and 1.

26. Differentiate between sigmoid and tanh.

Sigmoid range is 0 to 1; tanh range is -1 to 1.

27. What is softmax activation?

Converts outputs into probabilities.

28. In which applications is softmax used?

Multiclass classification problems.

29. What is the vanishing gradient problem?

Gradients become too small during training.

30. Which activation functions help reduce vanishing gradients?

ReLU and Leaky ReLU.

31. What is training in ANN?

Process of learning by updating weights.

32. What is an epoch?

One complete training cycle.

33. What is batch size?

Number of samples processed together.

34. What is iteration in ANN training?

One update step during training.

35. What is forward propagation?

Data moves from input to output layer.

36. What is backpropagation?

Weights are updated using error.

37. Why is backpropagation important?

It improves model accuracy.

38. Explain gradient descent.

Optimization method to reduce error.

39. What is learning rate?

Controls speed of learning.

40. What happens if the learning rate is too high or too low?

High causes instability; low causes slow learning.

41. What is loss function?

Measures prediction error.

42. Differentiate between loss function and cost function.

Loss is for one sample; cost is average loss.

43. What is Mean Squared Error (MSE)?

Average squared prediction error.

44. What is cross-entropy loss?

Loss function used for classification.

45. What is optimization in ANN?

Process of improving model performance.

46. Name different optimization algorithms.

SGD, Adam, RMSProp, Adagrad.

47. What is stochastic gradient descent?

Updates weights using one sample at a time.

48. Differentiate between batch gradient descent and stochastic gradient descent.

Batch uses full data; SGD uses one sample.

49. What is overfitting?

Model memorizes training data.

50. What is underfitting?

Model fails to learn patterns.

51. Why is data preprocessing important?

Improves model accuracy and performance.

52. What is normalization?

Scaling data between fixed ranges.

53. What is feature scaling?

Making feature values similar in range.

54. Why do we split data into training and testing datasets?

To evaluate model performance.

55. What is validation data?

Data used to tune model during training.

56. What is one-hot encoding?

Converts categorical data into binary vectors.

57. What type of data is suitable for ANN?

Numerical and structured data.

58. What are categorical and numerical features?

Categorical are labels; numerical are numbers.

59. Why is missing data problematic?

It reduces model accuracy.

60. How can ANN handle noisy data?

By learning important patterns from data.

61. Which libraries are commonly used for ANN implementation in Python?

TensorFlow, Keras, PyTorch, NumPy.

62. What is TensorFlow?

Google’s deep learning framework.

63. What is Keras?

High-level API for deep learning.

64. Differentiate between TensorFlow and Keras.

TensorFlow is backend; Keras is user-friendly API.

65. What is PyTorch?

Open-source deep learning framework.

66. How do you create a neural network model in Keras?

Using Sequential() and Dense layers.

67. What is the purpose of model compilation?

To define optimizer, loss, and metrics.

68. What parameters are specified during model compilation?

Optimizer, loss function, metrics.

69. What is model fitting?

Training the neural network.

70. Explain the use of Sequential() model in Keras.

It creates layer-by-layer neural networks.

71. What is the role of Dense layer?

Fully connected neural network layer.

72. How do you evaluate a trained model?

Using test data and accuracy.

73. What is the use of predict() function?

To predict output for new data.

74. What is accuracy in ANN?

Percentage of correct predictions.

75. How can you improve model accuracy?

By tuning parameters and increasing data.

76. Why is differentiation used in backpropagation?

To calculate gradients for weight updates.

77. What is chain rule in ANN?

Method to calculate derivatives layer-by-layer.

78. Explain weight update rule.

New weight = old weight + learning adjustment.

79. What is the role of derivatives in ANN learning?

They help reduce error.

80. Why should activation functions be differentiable?

To perform backpropagation.

81. Explain the ANN experiment you performed.

Implemented ANN models like perceptron, XOR, CNN, and backpropagation.

82. What dataset did you use in your practical?

MNIST and Breast Cancer datasets.

83. What preprocessing steps were applied?

Normalization and scaling.

84. Which activation function did you use and why?

ReLU and sigmoid for better learning.

85. How many hidden layers did your model contain?

Usually one or two hidden layers.

86. Which optimizer did you use?

Adam optimizer.

87. What learning rate was selected?

Mostly 0.1 or default Adam learning rate.

88. How did you measure performance?

Using accuracy and loss.

89. What accuracy did you achieve?

High accuracy around 90% or above.

90. What challenges did you face during implementation?

Overfitting and parameter tuning.

91. Mention real-life applications of ANN.

Image recognition, speech recognition, recommendation systems.

92. How is ANN used in image recognition?

It detects patterns and features in images.

93. How is ANN useful in medical diagnosis?

Helps predict diseases from medical data.

94. What is the role of ANN in speech recognition?

Converts speech into text.

95. How is ANN used in recommendation systems?

Suggests products or movies based on user behavior.

96. Can ANN be used for weather prediction?

Yes, ANN predicts weather patterns.

97. How is ANN applied in autonomous vehicles?

Used for object detection and decision-making.

98. What are the advantages of ANN?

Learns complex patterns and improves accuracy.

99. What are the disadvantages of ANN?

Needs large data and high computation.

100. Compare ANN with traditional programming approaches.

ANN learns from data; traditional programming follows fixed rules.